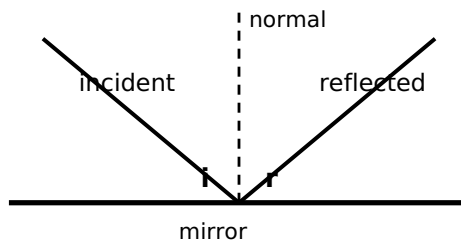


Laws of Reflection

Light bounces off any mirror by two simple rules — and a mirror's curve decides whether rays converge or diverge.

The two laws

The ray hitting the mirror is the **incident ray**; the one bouncing off is the **reflected ray**; the line at 90° to the mirror at the point of incidence is the **normal**.



Angle of incidence (i) = angle of reflection (r).

- 1 The **angle of incidence (i) = angle of reflection (r)**.
- 2 The incident ray, the normal, and the reflected ray all lie in the **same plane**.

These laws hold for **all** mirrors — plane, concave and convex.

Parallel rays on curved mirrors

- **Plane mirror:** parallel rays stay parallel after reflection.
- **Concave mirror:** parallel rays **converge** (meet) — can concentrate sunlight to burn paper.
- **Convex mirror:** parallel rays **diverge** (spread out).

Note: each ray still obeys $i = r$; the curve of the mirror is what makes the beam converge or diverge.

Where students lose marks

Trap 1 — Angles are measured from the NORMAL, not the mirror surface. i and r are between the ray and the normal.

Trap 2 — Concave converges, convex diverges. Don't swap them.

Trap 3 — If a ray hits along the normal, $i = r = 0^\circ$ and it reflects straight back.

Both laws apply to spherical mirrors too — the curve only changes the overall beam, not the rule per ray.

Model answers — write them like this

Example A. A ray hits a mirror at an angle of incidence of 35° . What is the angle of reflection?

Step 1: By the first law, $i = r$.

$\therefore$ Angle of reflection = 35° .

Example B. Why can a concave mirror burn paper in sunlight?

Step 1: It converges the parallel rays of the Sun to a point.

Step 2: The concentrated light gives enough heat to ignite the paper.

$\therefore$ Converging sunlight concentrates heat, burning the paper.

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